Example 5.15

ΔABC is equilateral and CD of the right triangle BCD is 8 *cm*. Find the side of the equilateral ΔABC and also BD.

Solution:

As ΔABC is equilateral from the figure, AB=BC=AC= $(x-2)$ *cm*.

$\therefore$ From ΔBCD, by Pythagoras theorem

$$BD^2 = BC^2 + CD^2$$

$$\Rightarrow (x+2)^2 = (x-2)^2 + 8^2$$

$$x^2 + 4x + 4 = x^2 - 4x + 4 + 8^2$$

$$\Rightarrow 8x = 8^2$$

$$\Rightarrow x = 8\,cm$$

$\therefore$ The side of the equilateral $\Delta ABC = 6$ *cm* and BD =10 *cm*.

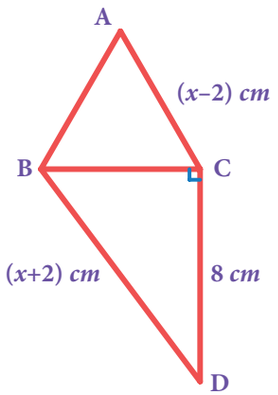


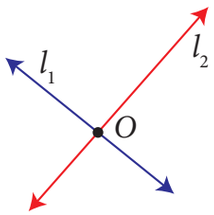
Fig. 5.28

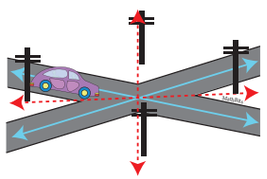
5.5 Point of Concurrency

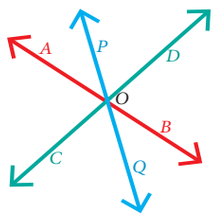
When two lines in a plane cross each other, they are called *intersecting lines*. Here, lines l_1 and l_2 intersect at point O and hence it is called the *point of intersection* of l_1 and l_2 .

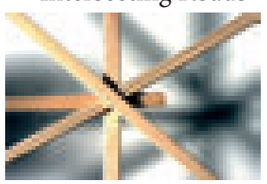
Three or more lines in a plane are said to be concurrent, if all of them pass through the same point.

In this figure, $\overleftrightarrow{AB}$, $\overleftrightarrow{CD}$ and $\overleftrightarrow{PQ}$ are concurrent lines and *O* is the *point of concurrency*.




Intersecting Roads




Concurring wooden pieces.

5.6 Medians of a Triangle

A median of a triangle is a line segment from a vertex to the midpoint of the side opposite that vertex.

In the figure $\overline{AM}$ is a median of ΔABC.

Are there any more medians for ΔABC ? Yes, since there are three vertices in a triangle, one can identify three medians in a triangle.

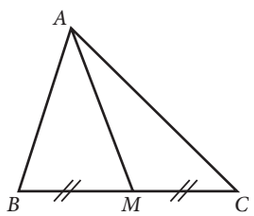


Fig. 5.29

Example 5.16

In the figure, ABC is a triangle and AM is one of its medians. If BM = 3.5 cm, find the length of the side BC.

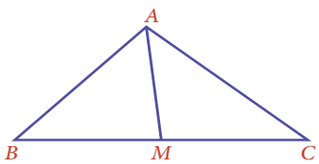


Fig. 5.30

Solution:

AM is median $\Rightarrow$ *M* is the midpoint of *BC*.

Given that, *BM* = 3.5 cm, hence *BC* = twice the length *BM* = 2 × 3.5 cm = 7 cm.

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